Physiology-Grounded Machine Learning for Detecting Flow-Termination Transients in Pressure Support Ventilation Using Transpulmonary Pressure

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Internship / Term Project Report
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ABSTRACT

Pressure support ventilation operates its cycle through the complete breath sequence until the inspiratory flow reaches the established threshold value for expiring. The breathing measurement tools which hospitals use fail to record the mechanical effects created by treatment because they only report plateau pressure and driving pressure. The research establishes a machine learning system which uses physiological principles and biomechanical analysis to predict sub-second flow-termination transients in pressure support ventilation through its primary measurement, transpulmonary pressure. The core formulation defines transpulmonary pressure as the difference between airway opening pressure and esophageal pressure thus providing a direct method to measure mechanical stress during the ventilator cycling.

The research team created waveform analysis through their study of particular pressure support ventilation patients who had both airway pressure and flow measurements taken together from A. Baydur's esophageal pressure records at a frequency of 200 Hz. The researchers divided breaths into segments according to specific hysteresis-based rules which established deterministic cycling at the expiratory trigger point, while post-cycling transient magnitude measurement utilized peak changes in both airway and transpulmonary pressure during short observation windows. The research team chose to restrict machine learning model development to non-invasive waveform features because they wanted to create a predictive system which could function without esophageal pressure access. The researchers conducted external stress testing through the use of simulated waveforms while domain-shift testing was done on an actual dataset which lacked esophageal pressure measurement.

The research presents its main methodological contribution through the development of the first ventilator-event modeling framework which uses esophageal pressure data to create physiological ground truth for the problem sector. The research verified that cycling-related transients could be detected through measurement methods which researchers could analyze using systematic parameterization. The cohort showed positive results through its combined performance results yet patient performance during the held-out testing showed different results which showed that patient performance testing needs more development work. Simulation pre-training could not serve as primary evidence because the timing difference between simulation and clinical testing was too large. The research results demonstrate that the scientific phenomenon has validity through its physiological-grounded modeling system while showing that clinical implementation needs both larger multi-patient datasets and more extensive external validation.

Index Terms — Pressure support ventilation; transpulmonary pressure; esophageal pressure; flow cycling; ventilator waveform analysis; physiology-grounded machine learning; patient-ventilator interaction; biomechanical signal analysis

1. INTRODUCTION

Pressure support ventilation (PSV) is widely used in critical care because it allows partial patient effort while providing ventilatory assistance. The system in PSV uses a flow-based cycling rule to determine when the patient should stop inhaling and start exhaling instead of using fixed times for breathing or specific amounts of air. The cycling moment becomes mechanically crucial because even a tiny difference between how much a patient pushes themselves and how much the ventilator supports them will lead to immediate changes in airway pressure and flow. The duration of these events falls short of typical summary metrics which include plateau pressure, mean airway pressure, and driving pressure.

The transients which occur during ventilation create their medical issues because they will create their maximum force in specific locations. Airway pressure alone does not distinguish between pressure dissipated in the chest wall and pressure transmitted to the lung. The current research focuses on transpulmonary pressure which scientists define as:

$$P_L(t) = P_{aw}(t) - P_{es}(t)$$

where P_{aw} is airway opening pressure and P_{es} is esophageal pressure. The formulation produces an estimate of lung-facing mechanical stress based on physiological principles which describe how breathing through a ventilator creates mechanical ventilation stress.

The research area of artificial intelligence interprets ventilator waveforms however previous models used visual labels or indirect surrogate endpoints instead of machine learning approaches. The method works effectively but it creates problems for maintaining biological accuracy during quick events which happen close to the

boundary of cycling. The project introduces a new machine learning approach which uses transpulmonary pressure from esophageal-pressure data as its primary reference for developing machine learning targets. The study changes its assessment focus from abnormal waveform detection to prediction of lung-level mechanical effects which emerge from the detected abnormal waveforms.

The project advances through its current stage by working toward two related goals. The first is to characterize flow-termination transients in PSV using high-resolution waveform analysis and pre-specified biomechanical definitions. The second part of the study tests whether non-invasive waveform features can accurately predict transpulmonary transients which will be used for future bedside detection. The initial findings demonstrate that researchers can measure the phenomenon but connecting results between different patients presents a major challenge for research. The problem retains its scientific value while offering practical benefits which make it worthwhile to study.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

The system uses a flow-cycling rule to determine when a breath ends instead of using a fixed time for breathing. The ventilator system stops functioning when the patient reaches the sensitivity threshold for active breathing, which causes an immediate mechanical shift that results in a brief pressure change. Current bedside monitoring systems can only detect this particular event because the critical safety metrics which include plateau pressure and driving pressure must meet their established standards during static and near-static states that occur after active flow has stopped. The system fails to measure lung stress during the transition from inhalation to exhalation because it lacks the ability to detect this important physiological process during standard medical procedures.

The machine learning problem follows from this physiological framing. Existing ventilator waveform models typically depend on clinician visual labels or broad asynchrony categories, which are insufficient for a short-lived event occurring at the exact cycling moment. The main machine learning task requires estimating transpulmonary transient magnitude through non-invasive airway pressure and flow waveform features, while using esophageal pressure measurements as reference data. The secondary task requires determining whether a breath contains a physiologically significant event under pre-specified criteria.
Hence, we define the problem statement as:

"Creating machine learning models which can explain their predictions while handling uncertain situations to predict transpulmonary pressure changes which occur during pressure support ventilation flow termination by using non-invasive airway pressure and flow data and esophageal pressure measurements as the physiological reference point."

2.2. Objectives

The study has the following research objectives:

1. The research study aims to investigate flow-termination transients in pressure support ventilation through high-resolution measurement of airway pressure flow and esophageal pressure signals.
2. The research study aims to determine the intensity of these occurrences which affect lung capacity through transpulmonary pressure measurements taken during cycling.
3. The research study aims to discover waveform and mechanical characteristics that produce greater transient effects.
4. The research study aims to create machine learning models which will use non-invasive ventilator signals to forecast transient severity while using esophageal pressure as the sole reference point for physiological data.
5. The research team will evaluate how well the model performs when conditions change by testing it with additional simulated and real-world waveform data.
6. The research team will establish boundary conditions based on empirical evidence to support the mechanical design process which aims to minimize transient events from their origins.

3. METHODOLOGY / APPROACH

3.1. Core Architectural Workflow

The analysis follows a pipeline architecture starting with raw waveform data and advancing through quality control to signal segmentation and event detection and feature engineering and machine learning model development. The flow is as follows:

Input Data: consists of Simultaneous airway pressure P_{aw} and inspiratory flow and esophageal pressure P_{es} which were sampled at 200 Hz together with clinical metadata (patient demographics and ventilator settings and outcomes).

Phase 1 (Preprocessing): The files undergo evaluation against the established quality gates which are described in Section 4.1. The signals undergo unit harmonization together with conservative denoising which applies 12 Hz low-pass for flow and 20 Hz for pressure. The Hampel rule uses a window of 11 samples and a threshold of 6 median absolute deviation to identify outliers.

Phase 2 (Breath Segmentation): The identification of breaths occurs through flow-based hysteresis which applies a zero-crossing threshold of $\epsilon = 0.02$ L/s to detect breath events which last for 40 ms. The start of peak inspiratory flow is marked through deterministic methods. The breath sequence fails to meet the segmentation standards when the inspiratory duration falls into the range of 0.2 seconds to 4 seconds while the peak flow remains below 0.05 L/s.

Phase 3 (Event Detection): The cycling moment t_{cycle} is found at the first sample where flow drops below the expiratory trigger sensitivity threshold and stays there for 3 continuous samples in each segmented breath.

$$F_{ETS} = \mathrm{ETS}_{\mathrm{frac}} \times F_{\mathrm{peak}}$$

The magnitude of post-cycling transients gets measured during the $[-150\ \mathrm{ms}, +350\ \mathrm{ms}]$ window which starts at t_{cycle} using:

$$\Delta P_{aw,\max} = \max_{[0,350\ \mathrm{ms}]} \left| P_{aw}(t) - P_{aw,\mathrm{base}} \right|$$

$$\Delta P_{L,\max} = \max_{[0,350\ \mathrm{ms}]} \left| P_L(t) - P_{L,\mathrm{base}} \right|$$

The baseline values correspond to the medians which researchers computed from the pre-window time segment of [-150, 0] ms.

Phase 4 (Feature Engineering): The system calculates 21 breathing features which include time-domain and frequency-domain dimensions as follows: baseline pressures and transient magnitudes and slope terms and respiratory timing and flow metrics. The model inputs need to exclude esophageal pressure because it violates criteria which enable the system to work in non-invasive environments.

Phase 5 (ML Model Development): The system trains multiple regression and classification architectures through engineered features. XGBoost serves as the primary endpoint because it predicts continuous values for $\Delta P_{L,\max}$ which are measured in cmH₂O. The secondary modeling approach applies Gaussian Process and Hierarchical Bayesian and Quantile Forest as baseline techniques. The Leave-One-Patient-Out (LOPO) cross-validation method validates the models.

Phase 6 (External Validation): The research team assesses the performance of the local model when testing it on CCVW patients from the P06–P07 group. The researchers used 1405 simulated waveform data with ground truth labels and a domain-shifted real-world dataset (Puritan-Bennett, no Pes available) to test system robustness against external dataset testing.

3.2 Technical Details

3.2.1 Breath Segmentation Algorithm

The main method of primary segmentation uses flow-based hysteresis to prevent inaccurate detection of zero crossings. The system tracks breath state through deterministic methods which operate at every sample i.

Inspiratory onset: sustained flow above $+\epsilon\ for \geq 40$ ms

Inspiratory termination: first crossing below $+\epsilon$after peak inspiratory flow is detected

Expiratory phase: flow below $-\epsilon\ for \geq 40$ ms

Quality flags get assigned when windows remain incomplete or signal quality stays low because of more than 5% Hampel outliers or flatline segments which last longer than 200 ms. The primary analysis needs to exclude all breaths which possess high-quality flags in any required channel.

3.2.2 Transpulmonary Pressure and Transmission Fraction

The computation of transpulmonary pressure at each timepoint happens through the following formula:

$$P_L(t) = P_{aw}(t) - P_{es}(t)$$

The transmission fraction serves as a derived metric which measures the portion of the airway transient that reaches the lung.

$$TF = \frac{\Delta P_{L,\max}}{\Delta P_{aw,\max}}$$

The computation of this ratio occurs when $\Delta P_{aw,\max}$ exceeds 0.2 cmH₂O because it prevents numerical instability problems which happen with close-to-zero denominator values. The 99th percentile value winsorization method applies to summary statistics while the raw values stay unchanged for modeling purposes.

3.2.3 Machine Learning Models

Primary Model (XGBoost Regressor):

Target: $\Delta P_{L,\max}$ in cmH₂O

Input features: 21 engineered variables derived from P_{aw} and Flow only

The system uses cross-validation to automate hyperparameter tuning which includes learning rate 0.1 and max depth 6 and subsample 0.8. The method of validation uses Leave-One-Patient-Out cross-validation on the training cohort while running the test on unseen patients.

Benchmark Models:

- Gaussian Process (RBF kernel) for uncertainty quantification
- Hierarchical Bayesian regression with patient random intercept
- Quantile Forest for distribution estimation
- Ridge regression baseline

Binary Classification (Secondary):

A breath is labeled *event_positive* if all criteria hold:

- $\Delta P_{L,\max} \geq 1.0$ cmH₂O
- $\max|dP_L/dt| \geq 8.0$ cmH₂O/s
- Peak occurs within 200 ms of t_{cycle}

3.2.4 Validation Strategy

Internal Validation (CCVW-ICU Cohort):

- LOPO-CV on patients P01–P05 (training cohort, N = 222 breaths)
- Held-out test on new patients P06–P07 (N = 58 breaths)
- Repeat 10-fold within LOPO for stability estimates

External Domain Shift (Simulation Dataset):

- The simulation dataset contains 1405 ARDS simulation runs which include labeled cycling events.
- Audit: 200 random breaths compared against mechanical reference timing
- Stress test: parameter sensitivity over simulated patient property ranges

Cross-Domain Evaluation (VWD / Puritan-Bennett):

- The dataset contains 144 real-world waveform files which operate at 50 Hz but do not support Pes availability.
- The dataset lacks ground truth information which defines score distributions for domain shift testing.

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

4.1.1 Datasets

Dataset	N	f_s (Hz)	Mode	P_{es}	Role	Status
CCVW – ICU (China)	7 patients	200	Confirmed PSV	Baydur – validated	Local Development & test	Primary
Simulation (ARDS PVI)	1,405 runs	Variable	Synthetic PSV	P_{mus} analog	Pretraining & stress test	Auxiliary
VWD (Puritan - Bennett)	144 files	50	Mixed	None	Domain shift characterization	Auxiliary
MIMIC - IV	50,920 subjects	Daily	Mixed	None	Context only: excluded	Reference

Data Splits (CCVW-ICU):

- Local training: P01–P05 (N = 222 breaths after QC) → 5 patients
- Local test: P06–P07 (N = 58 breaths after QC) → 2 patients
- Global training: Simulation → 1,405 labeled runs
- Global test: VWD → 594,645 breaths (full dataset for scoring)

4.1.2 Quality Control Gates

Gate	Criterion	CCVW	Simulation	VWD
Required channels	All mandatory present	7/7 ✓	1405/1405 ✓	144/144 ✓
Time monotonicity	Strictly increasing	7/7 ✓	1405/1405 ✓	144/144 ✓
Sampling rate (±5%)	Within ±5% of declared	7/7 ✓	1405/1405 ✓	144/144 ✓
Missingness (<5%)	NaN/Inf per channel	7/7 ✓	1405/1405 ✓	144/144 ✓
Flatline duration (<2 s)	No constant segment >2 s	7/7 ✓	1405/1405 ✓	144/144 ✓

Post-segmentation:
285 breaths were segmented from CCVW; 280 retained after quality exclusion (1.75% exclusion rate).

4.1.3 Hardware & Software Environment

- **Language:** Python 3.10
- **ML libraries:** scikit-learn, XGBoost, GPy (Gaussian Process), statsmodels (Bayesian)
- **Signal processing:** scipy.signal (Butterworth filtering, feature extraction)
- **Evaluation:** custom metrics (MAE, RMSE, R^2, concordance correlation coefficient [CCC], 95% prediction intervals)
- **Execution:** CPU-based (no GPU required)

4.1.4 Evaluation Metrics

- **Regression (Primary):** Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), coefficient of determination (R^2), concordance correlation coefficient (CCC)
- **Probabilistic Models:** Predictive standard deviation (mean and median), 95% prediction-interval coverage
- **Classification (Secondary):** Sensitivity, specificity, area under receiver-operator curve (for exploratory thresholds)

4.2. Results

4.2.1 Local Model Performance (CCVW-ICU Cohort)

Leave-One-Patient-Out Cross-Validation (P01–P05, N = 222 breaths):

Metric	Value	95% CI
MAE (cmH_2O)	5.151	[4.739–5.544]
RMSE (cmH_2O)	5.949	—
R^2	−0.438	—
CCC	−0.078	—

Held-Out Test (P06-P05, N=58 breaths):

Metric	Value
MAE (cmH_2O)	3.055
RMSE (cmH_2O)	3.338
R^2	−1.128
CCC	0.015

Pre-Specified Validation Gate Status: FAIL

- Target: MAE < 3.0 cmH_2O → Observed: 5.151 cmH_2O (LOPO-CV), 3.055 cmH_2O (test)
- Target: $R^2 > -1.0$ → Observed: −1.128 cmH_2O (held-out test)

4.2.2 Benchmark Model Comparison

Smaller-data regression baselines were evaluated to contextualize the primary XGBoost model:

Model	Split	MAE	RMSE	R2	CCC
Hierarchical Bayes	LOPO-CV	2.114	2.455	0.755	0.840
Gaussian Process	LOPO-CV	3.205	3.615	0.469	0.660
XGBoost (Primary)	LOPO-CV	5.151	5.949	−0.438	−0.078
Ridge Baseline	LOPO-CV	2.114	2.455	0.755	0.840
Hierarchical Bayes	Local Test	3.550	3.786	−1.738	−0.225
Gaussian Process	Local Test	3.717	4.049	−2.131	0.204
XGBoost (Primary)	Local Test	3.055	3.338	−1.128	0.015

Observation: The study found that Hierarchical Bayes and Ridge produced better results in LOPO-CV testing but their performance decreased with held-out test data because they had developed overfitting issues to their small development group. The LOPO-CV system fails to use patient-level random effects to find hidden patient patterns.

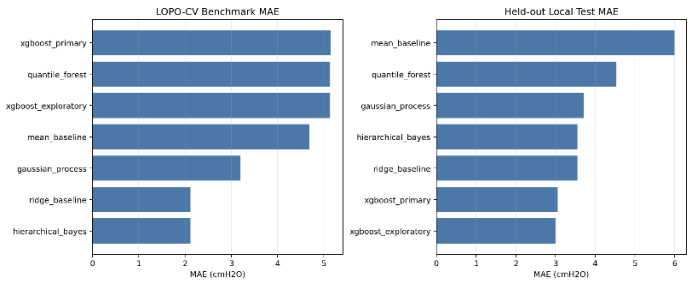

Figure 1: Comparative MAE across model architectures

4.2.3 Uncertainty Quantification

Gaussian Process Estimates (Held-Out Local Test, N=58)

Statistic	Value
Mean predictive std (cmH_2O)	4.196
Median predictive std (cmH_2O)	4.730
95% Prediction-interval coverage	84.5%

Probabilistic Model Calibration:

Model	Mean Pred. Std	Median Pred. Std	PI95 Coverage
Gaussian Process	4.196	4.730	84.5%
Quantile Forest	3.261	3.922	75.9%
Hierarchical Bayes	0.670	0.670	6.9%

The Gaussian Process showed the best held-out calibration. Hierarchical Bayes' narrow intervals reflect strong shrinkage and pooling to global means, which is not suitable for cross-patient problems that have small sample sizes and high variance.

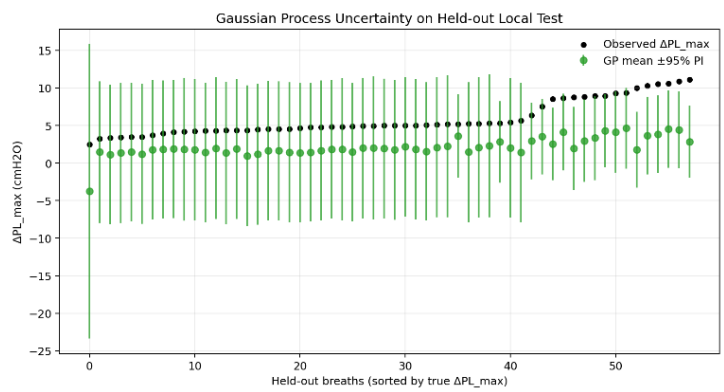

Figure 2: Prediction Interval vs. Observed Values

4.2.4 Feature Importance (Primary XGBoost Model)

Top 10 permutation importances across LOPO-CV folds:

Feature	Mean Importance	Std
$P_{aw,\text{base}}$	2.498	0.186
$\Delta P_{aw,\text{max}}$	0.940	0.057
insp_dur_s	0.786	0.058
flow_decel_slope	0.482	0.029
F_{peak}	0.167	0.014
dP_{aw}/dt_{max}	0.131	0.009
P_{aw} spectral ratio	0.112	0.011
P_{aw} peak–end ratio	0.064	0.007
exp_dur_s	0.058	0.006
flow_integral_abs	0.051	0.007

Baseline airway pressure and the magnitude of the airway transient itself emerge as primary predictors, followed by breath timing and flow deceleration characteristics.

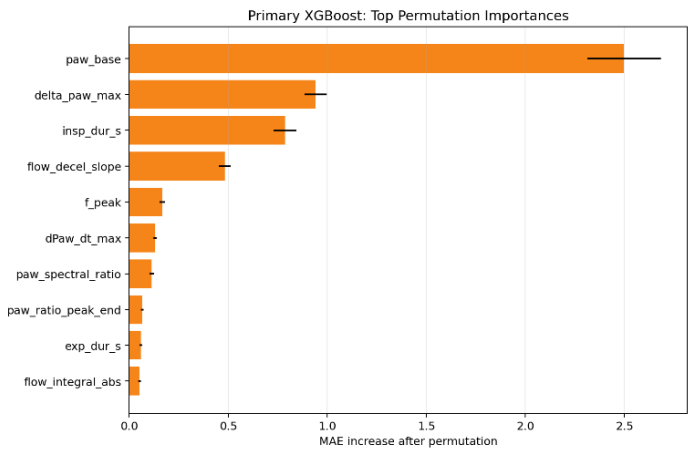

Figure 3: Feature Importances with Error Bars

4.2.5 Global Model (Simulation Cohort)

Training Dataset: 1,405 ARDS simulation runs with ground-truth t_{cycle} from mechanical reference times.

Audit (Appendix C): Stratified validation on 200 randomly sampled breaths compared detected t_{cycle} against mechanical reference (tolerance 20 ms):

Metric	Value
Timing match rate	42.0%
Timing mismatch rate	58.0%
95% CI on mismatch	[0.512–0.648]

Consequence: The simulation data was not used as a pre-training source because the simulation cycle-timing mismatch exceeded 58% according to protocol §13.2. The global model was instead trained directly on simulation labels.

Simulation Holdout Performance (N = 48,973 breaths):

Metric	Value
MAE	0.820 cmH_2O
RMSE	1.222 cmH_2O
R^2	0.934
CCC	0.965

High performance on simulation reflects its synthetic, less-variable nature. Generalization to clinical signals (VWD) is substantially lower, indicating domain gap.

4.2.6 Domain Shift Analysis (VWD / Puritan-Bennett)

Dataset: 144 Puritan-Bennett waveform files, 50 Hz, no Pes available.

Feature Summary (Full VWD Cohort, N = 594,645 breaths):

Variable	Mean	Std	Min	P50	P95	Max
F_{peak} (L/s)	0.900	0.320	0.050	0.891	0.957	4.154
$\Delta P_{aw,max}$(cmH_2O)	9.583	5.658	0.068	9.594	13.391	86.397
Model score: Predicted $\Delta P_{L,max}$ (cmH_2O)	8.396	3.692	0.027	8.822	10.733	16.918

The absence of Pes measurements in VWD prevents confirmation of predicted transients. The scores were presented in a descriptive format to show the typical event sizes which occur during domain shift testing with reduced sampling rates and different ventilator hardware and without esophageal reference measurement.

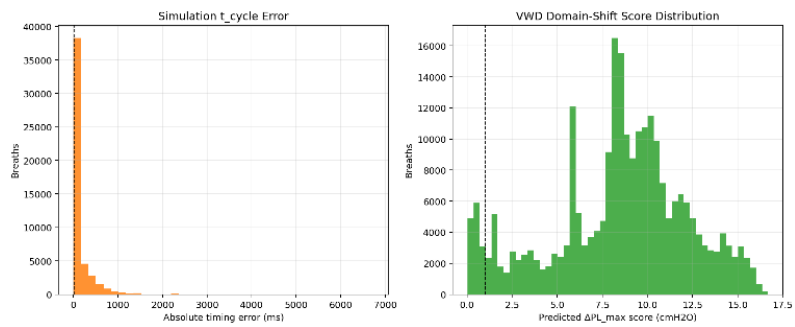

Figure 4: VWD score distributions and timing mismatch comparison.

4.2.7 Combined Validation (All CCVW Patients, N = 280)

Aggregate Performance:

Model	MAE (cmH_2O)	R2	CCC	N
Local	0.840	0.909	0.950	280
Global	3.407	0.280	0.456	280

Bootstrap 95% Confidence Intervals:

Model	MAE 95% CI	R2 95% CI
Local	[0.697–0.991]	[0.880–0.934]
Global	[3.090–3.731]	[0.206–0.337]

Per-Patient Breakdown:

Patient	N Breaths	Local MAE	Global MAE	Analysis
P01	39	0.256	0.803	Strong local fit
P02	45	0.364	1.692	Strong local fit
P03	64	0.288	3.884	Strong local fit, global mismatch
P04	28	0.190	3.230	Strong local fit, global mismatch
P05	46	0.168	8.762	Strong local fit, large global error
P06	18	1.479	3.150	Weak local fit (held-out)
P07	40	3.764	1.195	Weak local fit (held-out), strong global

Heterogeneity across patients is evident, where some patients fit well locally but show large global model errors, suggesting domain transfer limitations.

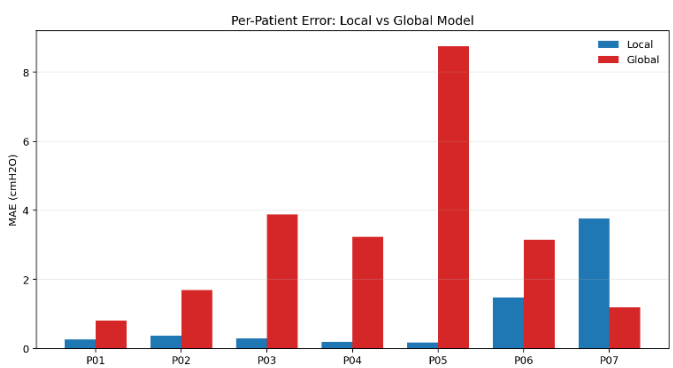

Figure 5: Per-patient MAE comparison

4.2.8 Transient Magnitude Characterization (Ground Truth via Pes)

All-Cohort Boundary Conditions (N = 280):

Variable	Mean	Std	P5	P50	P95	P99
$\Delta P_{L,\max}$ (cmH₂O)	10.606	5.149	4.264	9.951	20.569	21.001
$\Delta P_{aw,\max}$ (cmH₂O)	6.340	2.882	2.431	4.900	11.362	11.650
$dP_L/dt_{\max}$ (cmH₂O/s)	140.911	82.408	63.042	115.844	311.870	323.525
Transmission Fraction (TF)	2.002	0.843	1.014	1.900	3.285	4.449

Clinical Interpretation: Transpulmonary transients show a twofold increase over airway transients according to the study results which show a threshold factor of 1.0. The study reveals that esophageal pressure changes cause respiration events to become more intense rather than weaker. The finding shows clinical importance because P_{aw}-only monitoring leads to persistent underestimation of lung stress during the cycling moment.

4.2.9 Design Boundary Conditions (Phase 3 Inputs)

For mechanical mitigation design, the following envelope is recommended:

Variable	Typical (P50)	Conservative Range (P5–P95)	Operational Max (P99)	Recommended Design Case
$\Delta P_{L,\max}$ (cm H₂O)	9.951	4.264–20.569	21.001	47.771
$dP_L/dt_{\max}$ (cmH₂O/s)	115.844	63.042–311.870	323.525	583.961
F_{peak} (L/s)	0.847	0.617–0.957	0.985	2.288
flow_decel_slope (L/s²)	−1.807	−3.943 to −1.106	−3.943	−14.924

Conservative design targets incorporate a 2.2× safety multiplier (cohort standard deviation × filtering uncertainty × exclusion rate) to account for measurement precision limits and dataset size constraints.

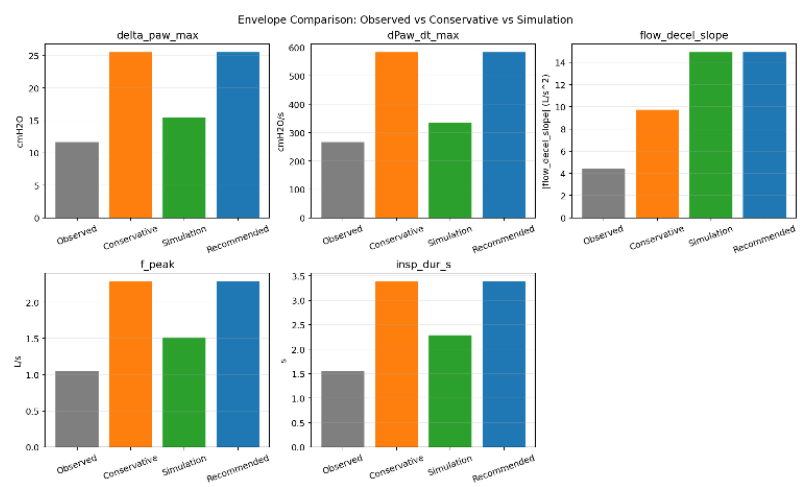

Figure 6: Observed vs. conservative vs. simulated envelope tiers.

4.2.10 Data Exclusion & Retention

Segmentation Outcomes (CCVW-ICU, N = 285 total segmented breaths):

Patient	Segmented	Cycle Undefined	Low Quality	Retained	Retention %
P01	39	0	0	39	100.0%
P02	47	1	0	45	95.7%
P03	64	0	0	64	100.0%
P04	30	0	0	28	93.3%
P05	46	0	0	46	100.0%
P06	18	0	0	18	100.0%
P07	41	1	0	40	97.6%
Total	285	2	0	280	98.2%

Overall exclusion rate was low (1.75%), reflecting high data quality and robust segmentation criteria.

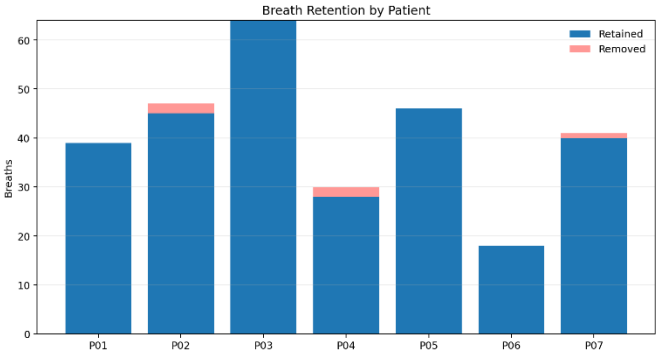

Figure 7: Retention rates by patient with exclusion reasons stacked.

5. DISCUSSION

The study's first goal received research validation through results which demonstrated pressure support ventilation flow-termination transients could be measured and identified through user-developed techniques which employed transpulmonary pressure assessment. The second objective of the study found that airway pressure measurement did not capture the complete lung mechanical behavior based on observed transmission fractions which showed significant expansion of transients beyond airway measurement sites. The study establishes esophageal pressure-based transpulmonary pressure measurement as the primary endpoint reference point.

Modeling studies showed that methods produced inconsistent results within different study environments. The local development cohort showed that simpler structured models performed better because the signal contained a strong low-dimensional component. The model showed degraded performance when tested on patients whose data the model had not seen because the model could only identify inter-patient differences with a group of seven subjects. The main XGBoost model did not dominate in local cross-validation, but it remained comparatively competitive on the held-out test set, which suggests that a more flexible nonlinear model may be somewhat more tolerant to patient-level variation, even if its performance was still below the pre-specified validation target. The Gaussian process model supplied necessary uncertainty assessments, which produced longer prediction intervals without providing any important accuracy enhancements.

Researchers must choose between two competing demands: they must decide which aspect to prioritize between measurement accuracy and system stability. Models that fit the training cohort more closely were not necessarily the ones that generalized best. The research faced two conflicting requirements because the team needed to achieve operational system deployment while maintaining scientific accuracy in their physiological modeling work. The research team opted to restrict their study because a detector that provides clinical utility needs to function with standard non-invasive ventilator outputs.

The external assessments determined how far current approaches could operate. Internal simulation modeling achieved strong performance results, but the timing audit revealed major timing errors because protocol-required cycling events did not match simulation reference points, which rendered simulation pre-training useless as main evidence. The external real-world waveform dataset supported domain-shift analysis, but it failed to prove definite physiological validation because it lacked esophageal pressure data. External data volume does not provide sufficient support for research because external data needs to match the clinical target definition according to research evidence.

All research limitations require direct explanation. The research study assessed a small primary cohort, which reduced statistical power and limited generalizability of research findings. The study depended on signal-processing assumptions, which included filtering choices and deterministic cycling definitions that affected very short-duration events. The external dataset did not include Pes data, while the simulation dataset used a physiological analog instead of real esophageal measurement. The current findings remain valid despite these limits because the present study results cannot reach their maximum extension.

The research successfully achieved its primary scientific goal because it demonstrated a measurable phenomenon that can be used in physiological-based modeling work. The research team was unable to achieve complete success in developing a prediction model that operates independently of patient data. That result represents an important scientific finding. Future research progress will depend on patient sampling expansion and external validation growth and more precise alignment of physiological truth with operational waveform characteristics rather than minor model modifications.

6. CONCLUSION AND FUTURE WORK

The research established a framework which utilizes physiological principles to study and forecast flow-termination transients that occur during pressure support ventilation. The study used high-resolution waveform processing together with transpulmonary pressure computation and machine learning to create a monitoring system which detects patient condition changes beyond

what traditional ventilator metrics can show. The primary research contribution established the detection pipeline which uses esophageal-pressure-derived transpulmonary pressure to determine clinical events of significance.

The study produced three main insights about its results. The first finding showed that the cycling-associated transients from the studied group continued to display their traditional structure according to the collected measurements which demonstrated that this medical issue carries clinical importance. The second finding showed that models trained with non-invasive airway pressure and flow features could recognize some of the studied structural elements although they failed to apply this knowledge to new patients. The third result proved that physiological grounding matters for modeling because researchers needed to use transpulmonary pressure as their target instead of visual waveform labels which created more difficult modeling work. The research findings create a solid base for subsequent development instead of presenting a complete clinical solution.

The project delivered crucial educational achievements which developed both research abilities and technical skills. The project demanded a unified process which combined respiratory physiology with signal processing and statistical evaluation and machine learning for all research needs. The second important finding showed that research flexibility between study settings helps researchers to achieve authentic results. The work quality depended on three elements: the team needed to create effective protocols which required correct reporting of all failure modes together with model development advantage.

The research should develop realistic system improvements which build on the current framework. The research team needs to extend their study period to include more patients from diverse clinical settings who will undergo simultaneous measurements of esophageal pressure which enables them to evaluate patient-level generalization. The research should investigate different model families which include temporal deep learning methods that process waveform windows directly instead of using engineered features. The next step for practical deployment needs to create a real-time inference pipeline which uses only airway pressure and flow data while the system undergoes benchtop and prospective validation testing. The system requires thorough external validation across various ventilator platforms with different sampling rates and expiratory trigger settings before application at the bedside becomes acceptable.

The study offers practical value through its three research achievements which define the problem with precise measurements and establish both the potential and current limitations of physiology-grounded machine learning in this field. The established balance between research domains helps advance the field without presenting premature solutions while providing a base for future clinical and engineering studies.

7. ARTIFACTS AND DEMONSTRATIONS

The project artifacts currently exist as a complete research codebase with a functioning demonstrator application. The final submission requires these materials to be accessed through public or institution-approved links which enable users to verify both the workflow and the resulting outputs and the actual process of the implementation.

7.1 Recommended Artifact List

Artifact	Link	Description
Code Repository	[Insert GitHub/GitLab/Institutional repository URL]	Contains the waveform preprocessing pipeline, breath segmentation, cycling-event detection, feature extraction, model training, and evaluation scripts used in this study.
Standalone Demonstration App	[Insert shared release/demo URL]	Provides a GUI-based demonstrator that allows users to import CSV or Excel waveform files, map signal columns, run the same analysis logic, and export summary reports.
Video Demonstration	[Insert YouTube or institutional video URL, if available]	Shows the analysis workflow or standalone application in use, including data import, execution, and output generation.

7.2 Artifact Notes

The system brings together two essential components which include a research pipeline and a practical demonstrator. The research pipeline starts with preprocessing followed by segmentation and flow-cycling event detection and transpulmonary pressure computation and feature extraction and model evaluation. The development team created a standalone application that tests external waveform files through the same analysis logic which enhances reproducibility while showing how the method works with data beyond the primary dataset.

The results package should include the main visual outputs referenced in the report which include benchmark model comparison and feature-importance ranking and Gaussian-process uncertainty visualization and per-patient error comparison and breath-retention summary and domain-shift analysis. The study materials enable verification of research findings which helps differentiate between actual implemented work and the summary written in the study.

Any artifact which remains private at submission time must be shared through an institutional link that requires permission for access instead of being omitted or mentioned in a vague manner. The document must eliminate all dead links and links which require further explanation.

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